MILITARY SPECIFICATION

FUZE, MT, SQ, M582
LOADING, ASSEMBLING AND PACKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing for the Fuze, Mechanical Time and Superquick, M582.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-P-60412 - Packaging, Packing and Marking for Shipment of Artillery Type and Rocket Fuzes, General Specification for

STANDARDS

MIL-STD-331 - Fuze and Fuze Components, Environmental Performance Tests for

DRAWINGS

PICATINNY ARSENAL

9236701 - Fuze, Mechanical Time and Superquick, M582

FSC: 1390

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3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Parts and Assemblies.-The parts and assemblies shall comply with all requirements specified on the drawings and with all requirements specified in applicable specifications.

3.3 Workmanship.-All parts shall be fabricated and finished in a thorough, workmanlike manner and all manufacturing, processing and assembly operations shall be correctly performed. The parts shall be clean and free of burrs, sharp edges, unblended radii, surface defects, cracks, chips, dirt, grease and oil (except where specifically required), rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat, legible and sharply defined.

3.4 Jolt and Jumble.-The fuze shall comply with the requirements of MIL-STD-331, Tests 101.1 and 102.1.

3.5 Transportation Vibration.-The fuze shall comply with the requirements of MIL-STD-331, Test 104.

3.6 Waterproofness.-The fuze assembly shall comply with the requirements of MIL-STD-331, Test 108.

3.7 Proving Grounds

3.7.1 Premature Bursts (see 6.4).-There shall be no premature bursts in the gun bore, in flight or during handling.

3.7.2 Functioning (see 6.5)

3.7.2.1 Target Impact Functioning.-The fuze, when set PD shall cause the projectile to function upon impact of the PD element when fired in the 155MM, M1 Tube, Charge 1 at a 2" plywood target located 820-5 feet from the muzzle of the weapon.

3.7.2.2 Airburst Functioning.-The fuze, when set for airburst functioning shall cause the projectile to function at or about its intended functioning zone along the projectile trajectory.

3.7.3 Target Non-Function.-The fuze when set PD shall not cause the projectile to function upon impact of the PD element when fired in the 105MM, M103 Tube, Charge 7 at a 2" plywood target located 150-5 feet from the muzzle of the weapon.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detailed specification.

4.2 Classification of Inspections.-The following types of inspections shall be conducted on this item:

a. Quality Conformance Inspection

4.3 Quality Conformance Inspection.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, each inspection lot of complete fuze assemblies shall contain:

a. Metal parts of not more than one lot from one manufacturer.
b. Booster pellets of not more than one lot from one manufacturer.

4.3.2 Examination.-See MIL-A-48078

a. Sampling Plans.-Unless otherwise specified in the Classification of Defects and Tests tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48880(PA)

PARAGRAPH	TITLE		SHEET 01 OF 01		DRAWING NUMBER 8595510
4.3.2.1	Pellet, Booster				NEXT HIGHER ASSEMBLY 9236701
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major A					
101	Pellet cracked, chipped or broken		0.40%	3.2	Visual
Major B					
131	Thickness		0.40%	3.2	Gage
132	Weight		0.40%	3.2	Gage
133	Outside diameter		0.40%	3.2	Gage
134	Radii		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.3	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48880(PA)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.3.2.2	Fuze, Mechanical Time and Superquick, M582, Less Booster	01	01	9236700
				NEXT HIGHER ASSEMBLY 9236701

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Fuze damaged		0.40%	3.2	Visual
102	Booster Cup damaged or missing		0.40%	3.2	Visual
103	Output lead functioned		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.0%	3.3	Visual
202	Marking Improper		0.65%	3.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48880(PA)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.3.2.3	Fuze, Mechanical Time and Superquick M582 prior to assembling Cup, Booster	01 OF 01	9236701
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Explosive material on booster cup threads		100%	3.2	Visual
Major A					
101	Booster pellet missing		0.40%	3.2	Visual
102	Foreign matter in booster cup		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.0%	3.3	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48880(PA)

PARAGRAPH	TITLE		SHEET 01 OF 01		DRAWING NUMBER 9236701
4.3.2.4	Fuze, Mechanical Time and Superquick, M582				NEXT HIGHER ASSEMBLY Final Assembly
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Length below Ogive, Max.		0.40%	3.2	Gage
102	Booster cup cracked or otherwise defective		0.40%	3.2	Visual
103	Adhesive missing or excessive		0.40%	3.2	Visual
104	Fuze not set in ship position		0.40%	3.2	Visual
105	Booster cup fails disassembly torque		0.40%	3.2	Gage
106	Booster pellet loose (moves when shaken)		0.40%	3.2	Visual/Manual
Major B					
131	Fuze threads damaged		0.65%	3.2	Visual
132	Fuze Ogive damaged		0.65%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.0%	3.3	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 01 OF 01		DRAWING NUMBER
4.3.2.5	Fuze, Mechanical Time and Superquick , M582 Destructive Tests				9236701
					NEXT HIGHER ASSEMBLY: Final Assembly
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Jolt and Jumble Test	6		3.4	4.4.1
2	Transporation Vibration Test	10		3.5	4.4.2
3	Target Non-Function Test	10(a)		3.7.3	4.4.4.3
4	Premature Burst	30		3.7.1	4.4.4
Major A					
101	Waterproofness Test	5		3.6	4.4.3
102	Target Function	10(a)		3.7.2.1	4.4.4.1
103	Airburst Function	10(a)		3.7.2.2	4.4.4.2

NOTES:

(a) These samples are part of thirty (30) required for the entire ballistic test.

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4.3.3 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.3.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and section 6 herein.

4.4 Test Method and Procedures

4.4.1 Jolt and Jumble Test.-The loaded fuze assemblies shall be subjected to the jolt test in accordance with MIL-STD-331, Test 101.1. After completion of the jolt test, the fuzes shall be subjected to the jumble test in accordance with MIL-STD-331, Test 102.1. Any fuze which does not comply with 3.4 shall be classed defective. One or more defects shall reject the lot.

4.4.2 Transportation Vibration Test.-The loaded fuze assemblies shall be subjected to the transportation vibration test in accordance with MIL-STD-331, Test 104, Procedure 2. Any fuze which does not comply with 3.5 shall be classed defective. One or more defects shall reject the lot.

4.4.3 Waterproofness Test.-The loaded fuze assemblies shall be subjected to the warerproofness test in accordance with MIL-STD-331, Test 108. After this test, the booster cup and pellet shall be disassembled and examined under ultra-violet light to determine compliance with 3.6. Any fuze which does not comply with 3.6, shall be classed defective. One or more defects shall reject the lot.

4.4.4 Ballistic Tests.Any premature burst during the entire ballistic test shall cause the lot to be rejected.

4.4.4.1 Target Functioning.-The fuze shall be assembled to a 155MM, M107 projectile and fired at an adequately supported target as specified in 3.7.2.1. When fired, the fuze, projectile and charge shall be at a temperature of 70° ± 10°F. This may require temperature conditioning in accordance with 4.4.5. Observation of impact functioning is required and shall be noted on the ballistic records. If more than one fuze fails to function the lot shall be rejected.

4.4.4.2 Airburst Functioning.-The fuze shall be assembled to a 105MM, M1 projectile and set for 25.0 seconds. The fuzed projectile shall be fired from a 105MM weapon with an M103 tube at Charge 7. When fired, the fuze, projectile and charge shall be at a temperature of 70° ± 10°F. This may require temperature conditioning in accordance with 4.4.5. All defects shall be recorded. A system of electric clocks or stopwatches shall be employed to measure the time to burst. These measurements shall be recorded to the nearest 0.01 seconds. If a round fails to function in flight observation shall be made to determine if the round functions on impact. If more than two (2) fuzes fail to function the lot shall be rejected.

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4.4.4.3 Target Non-Function Test.-The fuze shall be assembled to a 105MM, M1 projectile and fired at an adequately supported target as specified in 3.7.3. When fired, the fuze, projectile and charge shall be at a temperature of 70° ± 10°F. This may require temperature conditioning in accordance with 4.4.5. Observation of impact non-functioning is required and shall be noted on the ballistic records. If any fuze functions on target impact the lot shall be rejected.

4.4.5 Temperature Conditioning.-The projectile with the fuze assembled to it, and the charge shall be placed in a temperature conditioning chamber. With the components in the chamber, start the test at ambient temperature. The chamber shall then be maintained at the desired temperature until a steady-state condition exists. After a steady-state temperature condition is reached, the components shall be saturated for twenty-four (24) hours minimum at that temperature.

5. PREPARATION FOR DELIVERY

5.1 Packaging, Packing and Marking.-Packaging, packing and marking shall be in accordance with the requirements and inspection procedures specified in MIL-P-60412.

6. NOTES

6.1 Intended Use.-The components covered by this specification are intended for use in Artillery Systems.

6.2 Ordering Data.-See MIL-A-47078(PA).

6.3 Submission of Inspection Equipment Designs for Approval.-See MIL-A-48078. Submit equipment designs, as required to Commander, Picatinny Arsenal, ATTN: SARPA-QA-T, Dover, New Jersey 07801.

6.4 Premature Burst.-A premature burst is a malfunction in which a fuze functions before safe separation. In the point detonating mode (PD), safe separation is defined by the minimum arming distance per 3.7.3. In the airburst mode, safe separation is defined by the equation t-(4+.06t) where t is the set time.

6.5 Functioning.-A function is evidenced by the detonation of the specified projectile. A projectile which does not function as intended is termed a dud.

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6.5.1 Airburst Dud.-An airburst dud is a fuzed projectile set to function in the airburst mode but does not function in air. This is independent of whether or not the projectile functions upon subsequent impact.

6.5.2 Target Dud.-A target dud is a fuze which does not function on impact when it is intended.

6.6 Proving Ground Test Summary

TEST	WEAPON/ TUBE	CHARGE	DISTANCE/ SETTING	SAMPLE	ACCEPT	REJECT
Non-Function	105MM/M103	7	150-5 FT.	10	0	1
Function	155MM/M1	1	820-5 FT.	10	1	2
Airburst	105MM/M103	7	25.0 SEC.	10	2	3

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Preparing Activity
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Project Number: 1390-A-107

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